

# Learn HTML

## Lesson 5: Tables and Forms Basics

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### Tables and Forms Basics - Study Notes

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#### Key Concepts

1. **HTML Table Structure** – `<table>`, `<thead>`, `<tbody>`, `<tr>`, `<th>`, and `<td>` elements.
2. **Form Elements** – `<form>`, `<label>`, `<input>`, `<select>`, `<textarea>`, and `<button>` (or `<input type="submit">`).
3. **Associating Labels with Inputs** – using the `for` attribute on `<label>` and matching `id` on the input.
4. **Submitting Data** – the `action` and `method` attributes of `<form>` and the role of the submit button.
5. **Basic Accessibility & Semantics** – using proper table headings and label input pairing to improve usability for screen readers.

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#### Summary

Tables are a straightforward way to display tabular data on a web page. A table is built from rows (`<tr>`) that contain header cells (`<th>`) for column titles and data cells (`<td>`) for the actual content. Grouping rows inside `<thead>` (for headings) and `<tbody>` (for body data) adds semantic meaning and helps browsers and assistive technologies understand the structure.

Forms collect user input. The `<form>` element wraps all the interactive controls and defines where the data is sent (`action`) and how (`method`). Each input field should have a corresponding `<label>` that either wraps the input or uses the `for` attribute linked to the input's `id`. This pairing improves accessibility and makes the clickable area larger. A submit button—either `<button type="submit">` or `<input type="submit">`—triggers the form submission.

Together, tables and forms form the foundation for many everyday web pages: displaying schedules, product listings, contact information, and gathering user feedback.

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#### Important Points

- **Table markup hierarchy**:

```
```html
<table>
  <thead> ... </thead>
  <tbody> ... </tbody>
  <tfoot> ... </tfoot> <!-- optional -->
</table>
```
```

- Use `<th scope="col">` for column headers and `<th scope="row">` for row headers to aid screen

readers.

- Every `<input>` (except hidden) should have an associated `<label>`.
- The `for` attribute on `<label>` must exactly match the `id` of its input.
- Default form `method` is **GET**; use **POST** for sensitive data.
- The submit button can be created with:

```
```html
<button type="submit">Send</button>
<!-- or -->
<input type="submit" value="Send">
```
```

- Keep forms short and logical; group related fields with `<fieldset>` and `<legend>` (optional for beginners).

- Basic CSS can improve readability (e.g., `border`, `padding`, `width`), but not required for functional basics.

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## Examples

### 1b ã Simple Table – Weekly Schedule

```
```html
<table border="1">
  <thead>
    <tr>
      <th>Day</th>
      <th>Morning</th>
      <th>Afternoon</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">Monday</th>
      <td>Math</td>
      <td>Science</td>
    </tr>
    <tr>
      <th scope="row">Tuesday</th>
      <td>English</td>
      <td>History</td>
    </tr>
  </tbody>
</table>
```
```

**\*Explanation\*:** The `<thead>` holds column headings. Each row in `<tbody>` starts with a row header

(`<th scope="row">`) for the day, followed by data cells (`<td>`).

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## 2b ã Basic Contact Form

``html

```
<form action="/submit-contact" method="post">
  <label for="name">Name:</label><br>
  <input type="text" id="name" name="name" required><br><br>
  <label for="email">Email:</label><br>
  <input type="email" id="email" name="email" required><br><br>
  <label for="msg">Message:</label><br>
  <textarea id="msg" name="message" rows="4" cols="30"></textarea><br><br>
  <button type="submit">Send Message</button>
</form>
```

``

**\*Explanation\*:** Each input has a matching ``<label>`` (`for` != `id`). The form uses `method="post"` to send data securely to `/submit-contact`. The submit button triggers the request.

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## Quick Review

1. What tags are used to define a table row and a table header cell?
2. How do you associate a ``<label>`` with an ``<input>`` element?
3. Which form attribute determines where the data is sent, and which determines how it is sent?
4. What is the default `method` for a form if none is specified?
5. Why should you use ``<th scope="col">`` or ``<th scope="row">`` in tables?

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## Further Reading

- **HTML Table Accessibility** – using `caption`, `summary`, and ARIA roles.
- **Form Validation** – HTML5 attributes (`required`, `pattern`, `min`, `max`) and basic JavaScript validation.
- **Styling Tables and Forms** – introductory CSS for borders, spacing, and responsive layouts.
- **Advanced Form Controls** – ``<select>``, ``<radio>``, ``<checkbox>``, and file uploads.
- **Semantic HTML** – ``<fieldset>`` and ``<legend>`` for grouping related form fields.

